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MILITARY SPECIFICATION

FUEL SYSTEM, AIRCRAFT, DESIGN, PERFORMANCE, INSTALLATION, TESTING, AND DATA REQUIREMENTS, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers requirements for design, performance, installation, testing, and data requirements of the fuel system and fuel subsystems of aircraft powered by air-breathing engines. In conflicts between this document and referenced documents, this document governs.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

WW-T-700	Tube, Aluminum Alloy, Drawn, Seamless, General Specification for

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-P-5238	Pump, Centrifugal, Fuel Booster, Aircraft, General Specification for
MIL-T-5578	Tank, Fuel, Aircraft, Self-Sealing
MIL-P-5902	Purging Gas Systems, Aircraft Fuel Tank, Internal and External, General Specification for
MIL-T-6396	Tanks, Fuel, Oil, Water-Alcohol, Coolant Fluid, Aircraft, Non-Self Sealing, Removable, Internal
MIL-L-6503	Lighting Equipment, Aircraft, General Specification for Installation of
MIL-E-7080	Electric Equipment, Aircraft, Selection and Installation of
MIL-F-7179	Finishes and Coatings: General Specification for Protection of Aerospace Weapons Systems, Structures and Parts
MIL-T-7378	Tanks, Fuel, Aircraft, External, Auxiliary, Removable
MIL-C-7413	Couplings, Quick Disconnect, Automatic Shut-off
MIL-V-7899	Valves, Fuel Check, Low Pressure
MIL-G-7940	Gages, Liquid Quantity, Capacitor Type, Installation and Calibration of
MIL-P-8045	Plastic, Self-Sealing and Non-Self Sealing Tank Backing Material

FSC 2915

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MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-T-8606	Tubing, Steel, Corrosion-Resistant (18-8 Stabilized)
MIL-V-8608	Valves, Fuel Shutoff, Electrical Motor Operated
MIL-F-8615	Fuel Systems Components; General Specification for
MIL-S-8710	Strainer, Aircraft Fuel System, General Specification for
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-C-22263	Coupling, Fuel Line, Flexible, 125 PSI
MIL-V-25023	Valve, Fuel Drain, Self-Locking
MIL-M-25047	Markings and Exterior Finish Colors for Airplanes, Airplane Parts, and Missiles (Ballistic Missiles Excluded)
MIL-E-25499	Electrical Systems, Aircraft, Design and Installation of, General Specification for
MIL-H-25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-T-25783	Tank, Fuel, Aircraft and Missile, Non-Self-Sealing, High Temperature, Removable
MIL-A-25896	Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2-1/2 Inch Diameter
MIL-S-25980	Switch, Float, Aircraft Fuel Level, General Specification for
MIL-S-26390	Switches, Pressure, Aircraft Fuel
MIL-G-26988	Gage, Liquid Quantity, Capacitor Type, Transistorized, General Specification for
MIL-T-27422	Tank, Fuel, Crash-Resistant, Aircraft
MIL-R-27521	Receptacle, Flying Boom, Aerial Refueling
MIL-V-38003	Valves, Fuel Level Control, Fuel Tank, Aircraft, General Specification for
MIL-C-38300	Connectors, Electrical, Circular, Multicontact, High Environment, Quantitative Reliability, General Requirements for
MIL-C-38373	Cap, Fluid Tank Filler
MIL-S-38449	Signal Amplifier, Universal, Flying Boom Aerial Refueling

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipments)
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-1247	Identification of Pipe, Hose, and Tube Lines for Aircraft, Missile, and Space Systems
MS24354	Drogue Cone, Nozzle and Reception Coupling - Type MA-2 Flight Pressure - Refueling System, Assembly of
MS29520	Nozzle, Pressure Fuel Servicing Locking, Aircraft, Type D-1
MS29525	Cap, Pressure Fuel Servicing, Aircraft, 2-1/2-Inch Flush Type
MS29526	Cap - Pressure Fuel Servicing, Nominal 2-1/2 Inch Diameter, Non-Flush Type

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MS33514	Fitting End, Standard Dimensions for Flareless Tube Connection and Gasket Seal
MS33583	Tubing End - Double Flare, Standard Dimensions for
MS33584	Tubing End - Standard Dimensions for Flared
MS33586	Metals, Definition of Dissimilar
MS33611	Tube Bend Radii
MS33645	Receptacle, Installation, Fuel Nozzle Jumper, Aircraft to Servicing Hose

Air Force-Navy Aeronautical

AND10064	Fittings - Installation of Flared Tube, Straight Threaded Connectors

PUBLICATION

Air Force Systems Command Manual

AFSCM 80-1	Handbook of Instructions for Aircraft Design

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Aviation Administration

FAA Advisory Circular 25-3	Protection of Transport Aircraft Fuel Systems Against Lightning

(Application for copies should be addressed to the Federal Aviation Administration, Washington, D. C. 20553.)

3. REQUIREMENTS

3.1 Subsystems. The fuel system shall consist of the following subsystems:

a. Engine feed and transfer subsystem

b. Aircraft fuel tank subsystem

c. Pressurization subsystem

d. Purging subsystem

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e. Inerting subsystem

f. Fuel vent subsystem

g. Fuel quantity gaging subsystem

h. Ground refueling and defueling subsystem

i. Aerial refueling subsystem

j. Fuel jettison subsystem.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials used in the manufacture of fuel systems of Air Force aircraft shall be of high quality, suitable for the environment expected for the aircraft, and shall conform to applicable Government specifications or equivalent contractor specifications. The use of contractor specifications must be approved by the procuring activity. All materials shall be sufficiently resistant to applicable fluids specified in the weapon specification to assure satisfactory operation as defined herein.

3.3.1 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, atmospheric conditions, or wear likely to be encountered in transportation, storage, or during normal service life.

3.3.2 Dissimilar metals. Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion.

* 3.3.3 Magnesium, copper, and cadmium. Magnesium and copper, or alloys of either one, and cadmium plating shall not be used in parts that come into contact with fuel.

3.3.4 Protective treatment. When materials are used that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during storage or service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will chip, crack, abrade, peel, or scale with usage, age, or extremes of climatic and environmental conditions shall be prohibited.

3.4 Reliability. The reliability program for the fuel system, fuel subsystems, and components shall be integrated with the overall system reliability program and shall comply with the requirements of MIL-STD-785.

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3.5 Maintainability. The maintainability program for the fuel system, fuel subsystems, and components shall be integrated with the overall system maintainability program and shall comply with the requirements of MIL-STD-470.

3.6 Design and performance. The fuel system and all of its subsystems and components shall be designed in accordance with the requirements of this specification and those dictated by the operational environment and mission of the weapon system as described in the weapon system specification. The fuel system including all subsystems and components shall function satisfactorily under all possible design operating conditions for the aircraft.

3.6.1 Approval of design and development. As the development of the fuel system progresses, procuring activity engineering approval of the contractor efforts shall be obtained for each document that controls the design and development of the fuel system, and for each document that substantiates performance of the fuel system, its subsystems, and critical components. Procuring activity approval of the installed fuel system shall be obtained before acceptance of production aircraft. Adequate documentation to define and control the development of the fuel system shall consist of all data listed in the appendix of this specification.

* 3.6.2 General overall fuel system. The fuel system shall be designed for maximum simplicity consistent with weapon system requirements. The same piping and components may be used in more than one fuel subsystem. Fuel tanks shall not be located over crew or passenger compartments. There shall be no external leakage from the fuel system. There shall be no leakage during the "mated" condition while aerial or ground refueling.

3.6.2.1 Fuel. The fuel system design shall be suitable for use of any fuel specified in the Engine Model Specification and System Specification as primary, alternate, or emergency fuel. Degradation of performance with alternate or emergency fuel shall be specified.

3.6.2.2 Environmental

3.6.2.2.1 Temperature environment. The fuel system shall be designed to operate throughout the temperature range shown in table I.

3.6.2.2.2 Icing. The fuel system shall be so designed that icing does not adversely affect the fuel system components. Safety of flight of the aircraft shall not be jeopardized when operating with 0.75 cubic centimeter of free water per gallon of fuel in excess of saturation at 80°F. The fuel system shall be capable of continuous operation with fuel initially saturated with water at 80°F and cooled to any temperature down to and including -65°F. The fuel system shall operate under these conditions with fuel containing no anti-icing additive without draining or cleaning the system.

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TABLE I

TEMPERATURE ENVIRONMENT

	FUEL TEMPERATURE		AMBIENT AIR TEMPERATURE	
	LOW TEMP	HIGH TEMP	LOW TEMP	HIGH TEMP
CLASS I Fuel system which is not affected by fuel heating 1/	-65°F	+135°F	-65°F	+160°F
CLASS II Fuel system with aerodynamic heating or when fuel is used as a heat sink	-65°F	Max fuel temp as determined by Engineering Design Analysis (Item 20.1.a(2) of Appendix)	-65°F	Max ambient temp as determined by Engineering Design Analysis (Item 20.1.a(2) of Appendix)

1/ Due to location in hot compartments, the temperature requirement for some components in a Class I fuel system may exceed the temperatures specified. These components shall be identified and the maximum temperature specified.

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*3.6.2.3 Pressure. The fuel subsystems and components, excluding the vent subsystem, shall be designed for an operating pressure of 60 pounds per square inch (psi) and a proof pressure of 120 psi. At a burst pressure of 180 psi at maximum operating temperature there shall be no evidence of failure, malfunction, permanent distortion, or external leakage. The system and components shall be designed to function properly when ground or aerial refueled from tankers having a pressure and flow rate relationship depicted by the curves on figure 1. Fuel tanks shall be designed for a minimum of 1.5 safety factor in relation to positive or negative pressure capability.

*3.6.2.4 Components. Components shall comply with the applicable military specification. Fuel system components not covered by specific Government specifications shall conform to the applicable parts of MIL-F-8615. Model specifications shall be prepared by the contractor in accordance with the appendix of MIL-F-8615 for all fuel system components except standard hardware such as bolts, O-rings, etc.

3.6.2.5 Center of gravity control. Normal sequencing of fuel tanks shall be sufficient for maintaining center of gravity in the aircraft. It shall not be necessary to transfer fuel between fuel tanks for the express purpose of maintaining the aircraft center of gravity within limits except for supersonic aircraft.

3.6.2.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable in accordance with the requirements of MIL-I-8500. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.6.2.7 Emergency operation. The design and performance of the fuel system shall be such that no single equipment or component failure in the fuel system or the hydraulic, pneumatic, or electrical system supplying power to the fuel system will cause critical structural failure or prevent safe recovery of the aircraft.

3.6.2.8 Installation

3.6.2.8.1 Fuel lines

*3.6.2.8.1.1 Fuel line support and routing. Fuel lines shall be supported in accordance with table II. Rigid fuel lines shall be supported so that the lines will not deflect out of position as a result of internal pressure or as a result of aircraft maneuvers. Sufficient clearance shall be provided around fuel lines to prevent chafing of the lines with the aircraft structure or other lines and components. Unlined metal support clamps shall not be used on metal fuel lines. All fuel lines shall be routed in the most efficient manner to obtain maximum protection against battle damage. Where possible, fuel lines shall be routed through the fuel tanks to minimize fuel losses and fire resulting from combat damage or fitting failure. Access for maintenance and inspection of line joints of the fuel system shall be provided. Fuel lines which may be subjected to damage due to maintenance, cargo handling, and personnel traffic shall be protected.

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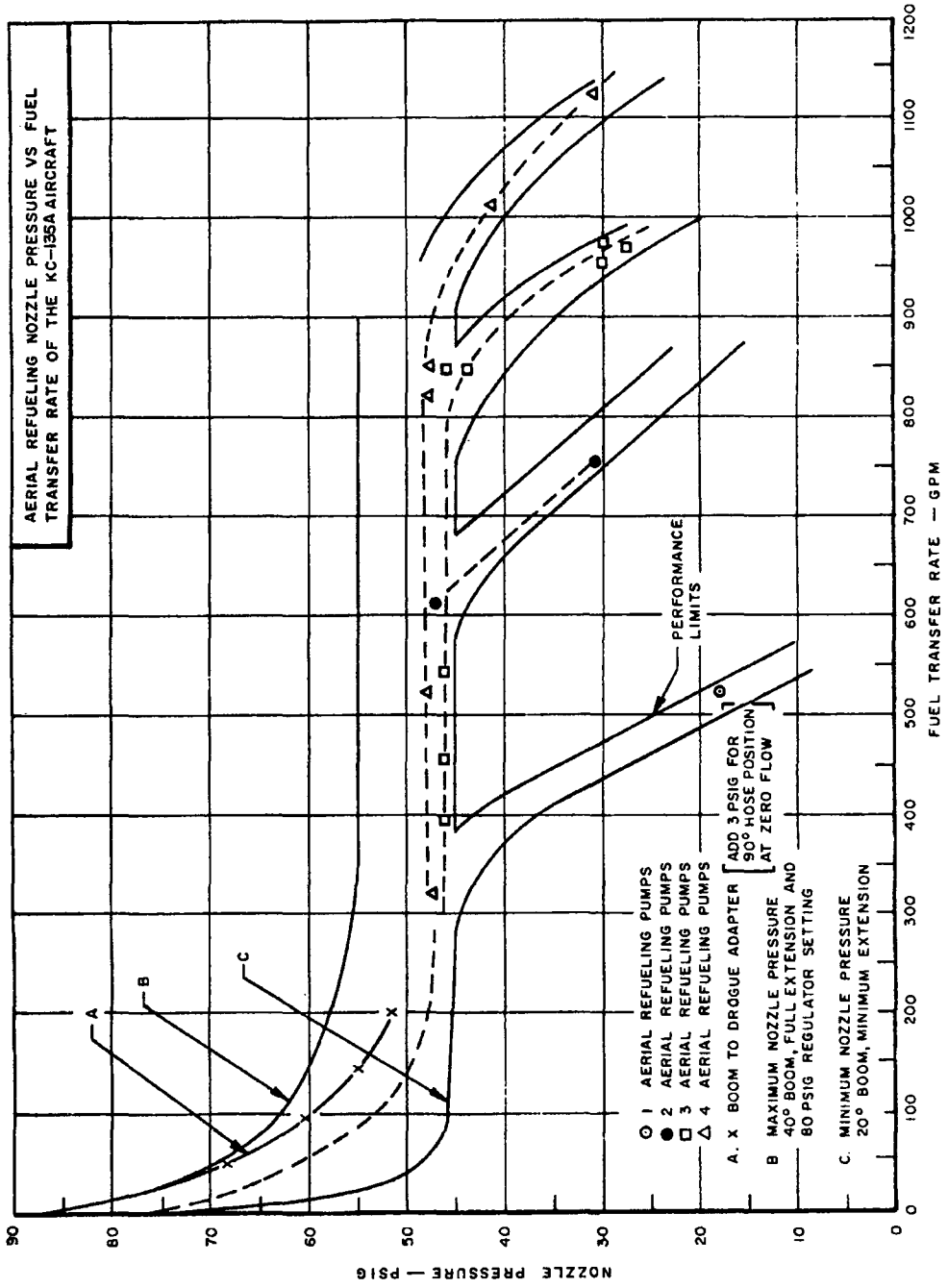


*FIGURE 1. Flow Curves

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TABLE II

FUEL LINES SUPPORT DISTANCE

Tube Size Inches	Maximum Distance Between Supports Inches
1 and smaller	24
1-1/4 to 1-3/4	27
2	31
2-1/2	33
3	34-1/2
4	37-1/2
5 and over	40

*3.6.2.8.1.2 Metal tubing. Metal tubing for fuel lines shall be fabricated from either stainless steel in accordance with MIL-T-8606 or aluminum alloy in accordance with WW-T-700 both of which are weldable. Fuel lines used in the fuel system shall have a wall thickness capable of withstanding an internal pressure of 480 psi without permanent deformation. The minimum wall thickness of tubing shall be .020 inch for stainless steel and .028 inch for aluminum. Aluminum alloy tubing shall be treated in accordance with MIL-F-7179. Welded tees and wyes are acceptable in the fuel system; however, extreme caution shall be used to obtain satisfactory welds. The welded tees and wyes shall not support any external weight or equipment items unless specifically designed for this purpose. Metal tubing shall be restricted to the following diameters: 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, 1-3/4, 2, 2-1/2, 3, 4, 5, 6, and 7 inches. Hose assemblies rather than metal tubing shall be used for 1/4-inch fuel lines except for drain lines. Minimum bend radii for metal tubing shall comply with MS33611. Forcing, bending, or stretching metal tubing to accomplish installation shall not be permitted.

3.6.2.8.1.3 Hose assemblies. Hose assemblies shall be in accordance with MIL-H-8795 or MIL-H-25579. Hose shall be installed with a bend radius no less than the minimum radius specified in the applicable military specification. Flexible hose connections when installed in the aircraft shall not be under tension, compression, or shear stress.

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3.6.2.8.1.4 Hot section fuel lines. Fuel lines shall not run through the hot section of the power plant compartment. If the routing of the fuel line through the hot section cannot possibly be avoided, then the line shall be stainless steel tubing with AND10064 connections. The procuring activity must approve the installation of fuel lines in the hot section.

*3.6.2.8.2 Fuel fittings. Bulkhead fittings in accordance with AND10064 shall be used as connectors for all tubing, hoses, or wiring where passing through the firewall and vaportight or liquidtight bulkheads. V-band-type connectors shall not be used in fuel lines. Self-sealing-type,quick-disconnect couplings used in the fuel system shall be in accordance with MIL-C-7413. Flexible couplings used in the fuel system shall be in accordance with MIL-C-22263. Tubing ends or fitting ends in accordance with MS33583, MS33584, or MS33514 shall be used with MS standard tube fittings in solid connections.

3.6.2.8.3 Aircraft accessories. All aircraft fuel system components shall be made accessible for inspection, cleaning, and adjustment or replacement while installed on the aircraft, with tools normally found in the mechanic's tool kit and without removal of the engine, fuel tanks, or important parts of the aircraft structure.

3.6.2.8.4 Bonding. The aircraft fuel system shall be electrically bonded in accordance with MIL-B-5087.

*3.6.2.8.5 Fuel drains. Fuel drains shall be incorporated in such a manner that no drainage impinges upon or re-enters the aircraft so that safe draining can be accomplished under all operating conditions. Fuel drains shall not be interconnected with component seal drains or with drain lines for electrical accessories draining oil, hydraulic fluid, water-alcohol, etc. Drain lines shall not be used for electrical wiring conduit. All nacelles, bladder tank cavities, and sheet metal pockets and traps where fuel may collect shall be drained to the exterior of the aircraft by a sufficient number of drains with 3/8-inch minimum diameter tubing or drain holes.

3.6.2.8.6 Marking. All fuel lines shall be identified by color bands in accordance with MIL-STD-1247.

3.6.2.8.7 Pipe threads. Tapered pipe threads shall not be used in the fuel system except for permanent closures.

3.6.2.9 Electrical equipment. All electrical equipment associated with the fuel system shall be in accordance with MIL-E-25499 and MIL-E-7080 except that connectors shall be in accordance with MIL-C-38300. Electrical equipment shall be isolated, where possible, from the fuel system to minimize the possibility of fuel leakage and fuel vapor coming in contact with electrical equipment through either the effect of gravity, airflow, or battle damage. Wiring shall be installed in accordance with MIL-W-5088. All electrical equipment shall meet the interference requirements of MIL-STD-826.

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3.6.2.10 Fire hazard reduction. Fuel lines shall not be run through any nacelle or power plant compartment except the ones they feed. The joints in fuel lines running through the cockpit shall be held to a minimum, and these joints shall be shrouded and drained to prevent fuel leakage from entering the cockpit.

3.6.2.11 Lightning hazards. The fuel system shall be designed in accordance with FAA Advisory Circular 25-3 to prevent fire and explosion when the aircraft encounters direct lightning strikes or conditions of crossfield streamering.

*3.6.2.12 Cockpit fuel control panel. The panel for the cockpit fuel controls shall have a simplified fuel system diagram inscribed on the panel. In this diagram the fuel tanks shall be represented by their respective quantity gages or by appropriate symbols. The flow control valves shall be represented by their respective control switches. The switches shall clearly indicate the ability of the fuel to either flow through the valve or be stopped by the valve. Critical switches shall incorporate safety guards. Multiple-position selector switches shall not be used for fuel system controls. Fuel valve controls shall be marked red and shall be grouped for convenient operation.

*3.7 Engine feed and transfer subsystem. It shall be possible to feed any engine from any tank and to cut off fuel flow to any engine or combination of engines without adversely affecting the fuel flow to the remaining engine(s). The fuel feed and transfer system shall be designed so that it is possible to consume the fuel in the main tanks without depending on boost pumps at altitudes up to 10,000 feet above sea level for fuel flow rates necessary for engine maximum power settings. Fuel from every tank shall be made available to the engines by suction feed at a fuel flow rate corresponding to cruise condition. External tanks are excluded from the suction feed requirements except where they serve as main tanks. The main tank for each engine shall be of sufficient capacity to contain fuel to meet the low level fuel requirements of 3.7.3. This is particularly important when a sump-type tank is used as a main fuel tank. On fighter-type aircraft the engine feed and transfer subsystem shall be completely automatic so that after the subsystem is activated all fuel aboard the aircraft will be delivered to the engine without further action by the pilot. Fuel from jettisonable external tanks with a capacity less than 1,000 gallons shall be transferred by pressurization of the tanks. For nonjettisonable tanks and tanks with a capacity greater than 1,000 gallons either pressurization or boost pumps may be used. The fuel system shall be capable of satisfactory performance during maximum rate of climb to the aircraft service ceiling, with each engine operating from its normal fuel supply, and with fuel in each tank at a temperature of 110°F at the start of the climb. This same performance is required with the fuel being pumped directly from the auxiliary tanks to the engine.

3.7.1 Engine inlet fuel feed conditions

3.7.1.1 Normal operation. The condition of the fuel, with regard to pressures, temperature, and all parameters which may affect engine operation, shall satisfy the conditions listed in the engine model specification under the most adverse conditions. The fuel system shall in no way limit aircraft performance within allowable engine and aircraft operating limits.

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3.7.1.2 Boost pumps inoperative. The vapor liquid (V/L) ratio of the fuel at the fuel inlet to the engine shall not exceed the V/L ratio specified by the engine model specification at altitudes up to 10,000 feet at military rated power with fuel boost pumps inoperative. This condition is determined for the worst possible operational condition of the feed system; that is, with the airframe fuel filter in the impending bypass condition at 110°F or the maximum fuel design temperature, whichever is higher.

3.7.2 Transfer system arrangement. The transfer system shall be so designed that the immediate attention or immediate action of the pilot (or flight engineer) during normal fuel system operation is not required for safety of flight. Transfer pumps shall be shut off automatically when the tank is empty on aircraft where the fuel system controls are not located on the flight engineer's panel. The transfer system shall deliver fuel to the main tank(s) or directly to the engines by an override system, at a rate not less than the rate of fuel consumption of the engine(s) in the military power condition for all altitudes.

3.7.3 Fuel availability warning. A low fuel level warning shall be provided in all aircraft. The low level warning shall indicate 20 minutes of fuel remaining at maximum range cruise power and altitude plus fuel for a normal descent and landing plus one missed approach.

3.7.4 Unavailable fuel. Unavailable fuel shall be held to a minimum consistent with sump capacity requirements. Tank outlets and boost pumps shall be so located that maximum main tank fuel is usable in landing approach and maximum auxiliary tank fuel is usable during level flight. Ejectors may be utilized to reduce the amount of unavailable fuel.

3.7.5 Extreme operating attitude. The aircraft shall be capable of uninterrupted fuel feed for a period of 3 minutes in any extreme operating attitude, excluding inverted flight, with the main tanks(s) 25 percent full. The airframe manufacturer shall specify the maximum allowable extreme attitude for pitch, yaw, and roll. A full continuous fuel flow shall be provided for the tank(s) to the engine(s) for at least 10 seconds during negative g for fighters, trainers, or any other aircraft required to fly in this manner.

3.7.6 Functional failure

3.7.6.1 Takeoff with failure. With any single failure of a functional component of the feed and transfer system, it shall be possible to attain 10,000 feet altitude and execute a safe return to base.

3.7.6.2 Failure during flight. No single failure of a functional component in the fuel system or other system necessary to the proper functioning of the fuel system shall result in an unsafe condition.

3.7.7 Feed and transfer surge requirement. Surge pressure shall not exceed 120 psi as a result of initiating or stopping intertank transfer or as a result of decreasing power settings of the engine(s).

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3.7.8 Feed and transfer subsystem components

3.7.8.1 Fuel boost pumps. Boost pumps shall conform to the requirements of MIL-P-5238. The boost pumps installed in fuel tanks of 1,000 gallons or greater capacity shall be removable without draining the tank. For pumps with three-phase, alternating-current motors which are installed for two-phase operation, the two-phase operation must be defined in the pump model specification approved by the procuring activity.

3.7.8.2 Fuel pressure switches. Each engine feedline shall contain a fuel pressure switch. Fuel pressure switches shall conform to the requirements of MIL-S-26390. Low fuel pressure warning lights shall be installed in the cockpit on the fuel control panel.

3.7.8.3 Airframe fuel strainers. The necessary strainers or filters shall be incorporated in the airframe fuel system to insure that the particle size of contaminants delivered to the engine is within the limits set forth in the applicable engine model specification. Airframe fuel strainers shall be in accordance with MIL-S-8710. The strainer shall be so located with respect to the fuel tanks and lines that the entire fuel system may be drained by opening the strainer drain, fuel tank sump drain, and booster pump drain plug. All fuel system strainers shall be equipped with bypass provisions to prevent engine fuel starvation by icing or plugging of the strainer element with contaminant. The bypass shall be designed to minimize washing ice or contaminant from the element or sump into the fuel system. An indicator shall be provided to show when the strainer is impending bypass. All fuel strainers shall be installed to be readily accessible, properly mounted by means of the mounting lugs provided, and with sufficient clearance to permit easy removal of the lower cover and element.

3.7.8.4 Fuel shutoff valves. Electrically operated shutoff valves shall conform to the requirements of MIL-V-8608. Electrically operated valves shall be so designed that there is no possibility of the valve changing position during flight due to a short in the aircraft electrical circuitry supplying power to the valve, electrical power failure, or normal mechanical forces. The powerline for each critical engine shutoff valve shall be connected to the essential bus with a minimum number of electrical components, such as plugs and relays, in the circuit. The critical valve wires shall be located as near the engine compartment as possible, but not in the compartment. The preferable location is on the outside face of the engine compartment firewall. Cockpit warning lights shall be provided for the firewall shutoff valves and fuel dump valves. These lights shall illuminate when the valve is in transition, a fault exists on the valve circuitry, or the valve limit switch fails to operate.

3.7.8.5 Check valves. Check valves shall be in accordance with MIL-V-7899.

3.8 Aircraft fuel tank subsystem. The fuel tanks on all aircraft shall be designed in conjunction with other fuel system components so that the fuel is continuously available to the engine under all design maneuvers of the aircraft. All fuel tanks shall be completely drainable while the aircraft is at ground attitude. The type of tankage shall be selected from table III after careful consideration of the aircraft mission relative to reliability, maintainability, passive defense features, and crash safety aspects.

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* TABLE III

SELECTION OF PROPULSION FLUID TANKS FOR AIRCRAFT

Type of Tankage	Design Criteria Document	Applications and Restrictions
Conventional Bladder Cells	MIL-T-6396	Select conventional bladder cells for application where protection against small arms or maximum crash resistance are not primary objectives and where fuel temperatures do not exceed 135°F or cell wall temperatures do not exceed 165°F.
Self-Sealing	MIL-T-5578	Utilize self-sealing tanks where protection against gunfire up to 20 MM is a requirement.
Nonintegral, Internal Nonflexible Tanks	MIL-T-6396	Use for special applications subject to approval of the procuring activity.
Crash Resistant Tanks	MIL-T-27422	Utilize crash resistant tanks for fuselage and wing tanks when primary mission of aircraft is personnel transport.
High Temperature Tanks	MIL-T-25783	Utilize high temperature tanks in high performance aircraft when fuel temperature exceeds 135°F or when cell wall temperatures exceed 165°F.
External Tanks	MIL-T-7378	Use for range extension or to meet ferry requirements.
Integral Tanks	AFSCM 80-1, Part B, Chapter 1, Section 7 and detailed requirements of Structural Work Statement	Use only for wing tanks and after careful consideration of other tanks listed above. Do not use integral tanks for center wing or fuselage installation or directly adjacent to engine or gun compartments on manned aircraft. Do not select integral tankage for aircraft of which the primary mission is ground attack, low altitude observation, or other missions which subject the aircraft to small arms fire. Before selecting integral tanks, consider carefully the problems of structural corrosion due to water collection and microorganism growth.

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3.8.1 Internal tanks. The initial layout of the aircraft shall determine the most advantageous shape of the fuel tank. Shallow or flat horizontal tanks with flat bottoms shall not be used except for thin-winged aircraft in which shallow fuel tanks are necessary.

3.8.1.1 Tank accessibility. Fuel tanks shall be designed to be accessible. All tanks except integral tanks shall be installed so that it is possible to remove the tanks without:

a. Removing any other part of the aircraft, except cowling or access panels

b. Dissassembling structural parts

c. Removing the engine or any part of the engine.

Ample clearance shall be provided for the removal of sumps or inspection plates.

* 3.8.1.2 Tank construction. Internal structure within the tank, such as baffles, flapper valves, and vent lines shall be designed and constructed to be as simple and reliable as possible. The very minimum of such internal devices shall be utilized to comply with the fuel system performance requirements. The number of cells and the number of fittings required per cell shall be held to a minimum. The fittings shall be located above the fuel level wherever possible without any sacrifice of performance. "O" rings or "mold-in-groove" seals shall be used for sealing openings. The contractor shall determine the maximum positive and negative pressure that each tank must withstand. This shall include the pressures resulting from all maneuvers and failure conditions. The tanks shall be designed with a minimum safety factor of 1.5 for these conditions. Tank design and construction shall be in accordance with the applicable specification of table III.

* 3.8.1.3 Expansion space. An expansion space to prevent tank overflow shall be provided in each of the fuel tanks equal to or greater than 3 percent of the total fuel volume of the tank with the aircraft in ground refueling attitude. Where two or more cells are interconnected to serve as a single tank, expansion space may be provided for the cell cluster instead of for the individual cells.

3.8.1.4 Sumps. Each fuel tank shall be provided with a sump (which shall be the lowest portion of the tank when the aircraft is at ground attitude) for the purpose of collecting sediment and water.

3.8.1.5 Sump capacity. The sump shall have a capacity equivalent to 0.0025 gallon per gallon (.25 percent) of tank volume to contain the water released from solution with the fuel in addition to that condensed in the fuel cells. The capacity of the sump shall never be less than 1 pint.

3.8.1.6 Sump drains. Sump drain valves shall be in accordance with MIL-V-25023. Drain bosses provided on booster pumps shall not be used as fuel tank sump drains.

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3.8.1.7 Access doors. Each cell shall be provided with an access door of such size that the entire interior of the tank is available for inspection and cleaning. Bolts smaller than 1/4 inch in diameter shall not be used for access doors or for installing components on the tank.

3.8.1.8 Outlet. The fuel tank outlet shall be located in such a position that water cannot collect and freeze around the impeller of the booster pump in cold weather. The fuel tank outlet fitting shall be so located as to reduce the unusable fuel to a minimum compatible with the sump capacity requirements of 3.8.1.5.

3.8.1.9 Tank accessories. Tank accessories, such as pumps, sumps, gages, vent lines, and outlet lines, shall be mounted so that their load is carried by aircraft structure.

* 3.8.1.10 Filler openings. The filler opening shall be so located to make it impossible to fill the expansion space. The filler openings shall be located to permit filling the tanks from outside the aircraft when it is on the ground without the use of special funnels. Filler openings shall also be located so that the liquids from the openings do not spill on parts of the aircraft structure, engine, or accessories, and thus cause damage or a hazardous condition. If necessary, filler spill basins shall be provided with adequate drains to prevent spilled liquid from entering the fuselage, engine compartment, and wings. The tank capacity, type fuel, and alternate fuel shall be stenciled adjacent to the filler opening in accordance with MIL-M-25047. The tank shall be properly vented so that when the filler cap is removed, complete filling can be accomplished in only one operation. Screens shall not be located in the filler neck. An internal guard shall be provided for thin-wing aircraft having shallow fuel tanks to prevent the fuel nozzle from striking and damaging the bottom of the tank. Such a guard shall not reduce the required filling rate. Filler caps shall be in accordance with MIL-C-38373. The mounting of the cap adapter to the airframe shall be capable of meeting the same lightning strike requirements as required of the filler cap.

3.8.1.11 Tank identification. Removable tanks shall be marked for identification in accordance with the applicable specification of table III.

* 3.8.1.12 Fuel tank installation

* 3.8.1.12.1 Tank installation time. The time required to completely remove and install a new flexible internal (self-sealing or nonself-sealing) fuel cell in an aircraft of four or more engines shall be 8 hours maximum; on aircraft with less than four engines the time shall be 4 hours maximum.

3.8.1.12.2 Rigid internal removable tanks. Rigid internal removeable fuel tanks shall be supported in padded cradles or on padded supports suitable for distributing the weight equally throughout the tank structure. Metal fuel tanks shall not be mounted rigidly to the aircraft. The padding shall consist of gasoline- and oil-resistant material which will not absorb liquids. Felt-type padding shall not be used. Means shall be provided for holding the tank in place under adverse loading conditions. Brackets or lugs attached to the tank walls shall not be used for supporting the tank.

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*3.8.1.12.3 Self-sealing tanks. Metallic fuel lines shall not be installed in contact with the walls of a self-sealing tank. Aircraft structure surrounding or supporting self-sealing tanks shall be designed and constructed to withstand all flight conditions and forces produced by the passage of a projectile through the confined liquid. All external self-sealing surfaces of a flexible self-sealing tank shall be protected by using backing board conforming to MIL-P-8045. The structure surrounding self-sealing tanks shall be liquidtight except for drain lines which shall be a minimum of 0.5 inch in diameter.

*3.8.1.12.4 Bladder tanks. Nonself-sealing bladder tanks shall be installed in cavities free from sharp projections or abrasive surfaces. Cavities shall be free of all metal chips or other foreign material that might damage the bladder. The structure surrounding bladder tanks shall be liquidtight except for drain lines which shall be a minimum of 3/8 inch in diameter.

*3.8.1.12.5 Integral tanks. Integral fuel tanks shall be designed and tested in accordance with structures requirements since integral tanks are not a fuel system component.

3.8.2 External tanks

3.8.2.1 Construction. External fuel tanks shall be constructed in accordance with MIL-T-7378 and designed for simplicity, low cost, and ease of mass production. For jettisonable tanks, fuel and vent valves shall be located in the aircraft, not in the tanks. See 3.7, 3.10, and 3.11 for transfer, vent, and fuel gaging requirements. The tanks shall be designed so that loads are carried primarily in the tank skin and utilize a minimum number of structural bulkheads. Lap joints or continuous seam welding shall not be used in joining segments of tank skins. Spot or seam welds shall not be used to attach seal rings or bulkheads to the tank skin. Sealing compounds shall not be used. Rivets shall not be used through the tank skin, access doors, or other tank components where sealing is required.

3.8.2.2 External tank installation. External fuel tanks shall be designed and installed in accordance with MIL-T-7378. The tank installation shall provide for the following:

a. A mechanical and electrical release, when jettisoning is required

b. Approved moistureproof electrical connection

c. Approved self-sealing quick disconnects for fuel and air pressure lines

d. Hinges and quick-release-type locking provisions for all pylon doors and access covers necessary for tank installation or removal.

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3.8.2.2.1 Ground clearance. External tanks shall be so designed that there will be a minimum of 3 inches clearance on a smooth runway or, if the aircraft is designed for rough runway operation, a minimum of 6 inches shall be provided with strut compressed and one tire flat for either landing or takeoff configuration combined with a corrugated landing strip.

3.8.2.2.2 Installation time. The time required to install the external tank shall not exceed 15 minutes.

3.9 Pressurization, purging, and inerting subsystems

3.9.1 Pressurization. The pressurization subsystem shall provide for pressure relief to prevent overpressurization of the tank as a result of a pressure regulator failure. The system shall provide for automatic venting to atmosphere whenever the pressurizing source is inoperative or shut off.

3.9.2 Purging. Purging systems installed shall be in accordance with MIL-P-5902.

3.9.3 Inerting. A fuel inerting system shall be provided for aircraft that operate in a speed region where the fuel tank wall temperatures exceed the spontaneous ignition temperature of the fuel.

3.9.3.1 Capacity. The system capacity shall be adequate to supply inerting capability for the maximum duration of flight including a 20-percent margin for safety.

3.9.3.2 Ullage. At no time during flight shall the oxygen in the ullage volume exceed 5 percent 0_2 by weight in a gas-inerting-type system.

3.10 Fuel vent subsystems. The subsystem shall be designed to prevent siphoning and spillage of fuel and to protect the fuel tanks from destructive pressures. Fuel spillage through the vent line during any maneuver shall not exceed 0.05 percent of the fuel from any one tank or 1 gallon, whichever is smaller.

3.10.1 Vent line routing. With the aircraft at ground attitude, the vent line shall be routed from the highest point within the tank, transversing the three dimensions of the tank and terminating in a suitable location outside of the aircraft. Siphon breakers shall be installed in the vent lines if necessary to prevent siphoning.

3.10.2 Vent line size. The vent line size shall be selected by the length of the vent line and the amount of air or vapor that must be handled. The volume of air or vapor that must be handled shall depend on the fuel consumption of the engine, size of the fuel tank, rate of refueling, type of fuel being used, rate of climb, rate of dive, and service ceiling of the aircraft. The vent lines shall be of adequate size to prevent tank overpressurization in the event of a level control valve failure during refueling. For refueling, it shall be assumed that each tank is filled individually with a refueling static pressure of 55 psi at the inlet to the refueling nozzle. The diameter of vent lines shall be a minimum of 1/2 inch.

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3.10.3 Vent outlet location. The tank vent outlet shall be located so that liquids and vapors emitted will not blow back into the aircraft or come in contact with the exhaust system, and foreign matter thrown up by the wheels of the aircraft will not enter or block the vent outlet. The outlet shall be located so that any moisture collecting in the line will drain to the outlet and will not permit the entrance of water into the vent system. The vent outlet location and configuration shall prevent the formation of ice which will block the outlet.

* 3.10.4 Interconnected vents. The fuel tank vent systems shall be designed so that fuel cannot be transferred through one tank vent to another tank. Fuel tank vents may be terminated at the same outlet location in the aircraft.

3.10.5 Vent pressures. For flexible nonmetallic and light-walled metallic fuel cells, the vent outlet terminal shall be designed and located so that the pressure in the vented space of the fuel tanks is never below ambient pressure. For bladder-type tanks, the cell cavity pressure shall never be greater than the pressure within the tank. Internal fuel tanks with structural metal walls may have negative pressure within the structural limits of the tanks.

* 3.10.5.1 Pressure checking. For aircraft where the vent system must be blocked during maintenance to pressure check the vent system or tanks, special aerospace ground equipment (AGE) shall be designed for the plugging of the vent lines. The design of the AGE shall not cause damage to the tanks in the event the AGE is left in the vent system at the conclusion of the maintenance. The AGE shall have a streamer or marker to prevent leaving the AGE installed.

3.10.6 External tank vents

3.10.6.1 Vent installation. An approved fuel and pressure vent system with self-sealing quick disconnects shall be installed for all jettisonable fuel tanks. The tank shall be capable of automatic venting to atmosphere whenever the pressurizing source is shut off, and this venting equipment shall be provided in the aircraft, not in the tank.

3.11 Fuel quantity gaging subsystem. All of the internal and external fuel shall be gaged.

3.11.1 Fuel quantity gaging subsystem components. All of the components of the fuel quantity gaging subsystem shall meet the requirements of MIL-G-26988.

3.11.2 Fuel quantity gaging subsystem installation. The fuel quantity gaging subsystem shall be installed in accordance with the requirements of MIL-G-7940.

* 3.11.3 Fuel quantity display. On single- and dual-engine aircraft a single indicator shall be provided to show only total fuel aboard the aircraft. Additional indicators or one additional indicator with a switching arrangement shall be provided to indicate fuel in other tanks including external tanks. When the switching arrangement is not

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activated, the indicator shall show the fuel remaining in the tank(s) directly feeding the engine. It shall not require in excess of 5 seconds to obtain a reading when using the switching arrangement. On aircraft equipped with three or more engines, one fuel quantity indicator shall be provided for totalizing all the fuel carried aboard the aircraft. Additional indicators shall be provided to show the fuel quantity of individual tanks.

3.11.4 Contractor specification. In the event that the requirements of MIL-G-26988 and MIL-G-7940 are not stringent enough or do not adequately define peculiar installation requirements, the contractor may prepare a component or installation specification as required. This specification shall clearly indicate the differences between it and MIL-G-26988 and MIL-G-7940. In no case shall a contractor specification be prepared wherein less stringent requirements are quoted. The contractor specification shall be approved by the procuring activity.

*3.12 Ground refueling and defueling subsystem. It shall be possible to refuel the aircraft to capacity without the application of external power. Provisions shall be incorporated for the capability of selecting any individual fuel tank for filling and conversely to avoid filling any tank. The fuel line between the main tank(s) and engine(s) shall not be subjected to the pressures imposed on the refueling manifold during refueling. There shall be no external leakage from the fuel system during or after refueling.

3.12.1 Pressure refueling. All aircraft with an internal fuel capacity of 600 gallons or more shall incorporate a pressure refueling system.

*3.12.1.1 Fuel servicing adapter. Fuel servicing adapters in accordance with MIL-A-25896 shall be provided for refueling the complete aircraft. The number of adapters shall be in accordance with the requirements of table IV. These adapters shall be located on the aircraft so that the use of ladders, supports, or elevating devices are not required for insertion of the nozzle. The type fuel and alternate fuel shall be stenciled adjacent to the refueling adapter in accordance with MIL-M-25047. The adapter shall be located so that damage to the adapter will be minimized in the event of a wheels-up landing.

3.12.1.1.1 Adapter grounding. All fuel servicing adapters shall be positively grounded to the airframe. An electrical ground receptacle shall be installed for grounding the fuel servicing nozzle. The ground receptacle shall be in accordance with MS33645 except that the receptacle shall be located:

a. Not more than 20 inches or less than 5 inches from the servicing adapter

b. Not closer than 12 inches from fuel tank vents or openings.

3.12.1.1.2 Adapter cap. Fuel servicing adapters shall be provided with MS29525 caps for flush-type installations or MS29526 caps for nonflush-type installations.

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* TABLE IV

PRESSURE REFUELING TIME

TOTAL AIRCRAFT FUEL WEIGHT (POUNDS)	GROUND REFUELING		MAXIMUM AERIAL REFUELING TIME 1/ (MINUTES) (Time may be interpolated between divisions) 2/
	MAXIMUM NUMBER OF SERVICING ADAPTERS PERMITTED	MAXIMUM TIME 1/ (Minutes) 2/	
450,000	4	40	50
350,000	4	32	40
250,000	2	32	30
100,000	2	20	15
50,000	1	18	10
25,000	1	8	7
10,000	1	5	4
3,900	1	2.25	2.25

1/ Based on refueling 90 percent total aircraft fuel (See 3.12.1.2b)

2/ Actual flow time

3.12.1.1.3 Adapter installation. The adapter shall be installed to accommodate an MS29520 ground servicing nozzle. The adapter installation shall be designed so that the nozzle has ample clearance for installation and operation by personnel wearing heavy gloves. There shall be no switch or valve actuators installed on the adapter which will not completely clear the nozzle envelope when the nozzle is connected.

*3.12.1.2 Refueling and defueling lines. Fuel lines shall be sized so that all tanks theoretically reach the full position simultaneously when refueling. For aircraft incorporating both ground and aerial refueling subsystems, the refueling line sizes shall satisfy the flow requirements of both systems. The line sizes shall be capable

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of handling flow schedules for pressure refueling as specified in table IV. The following criteria shall be used when designing to meet the ground refueling schedules of table IV:

a. It shall be assumed that the refueling system is capable of delivering 600 gallons per minute (gpm) for each servicing adapter at pressures of 45 pounds per square inch gage (psig) measured at the gage point of the refueling nozzle. In addition, the refueling system shall control the pressure measured at the gage point of the refueling nozzle to a maximum pressure of 55 psig for low flow rates.

b. It shall be assumed that all tanks except main tanks are empty and that 10 percent of the total aircraft fuel capacity is distributed equally in the main tank(s) at the start of the refueling operation.

All refueling lines shall have thermal relief provisions. All fuel trapped in the refueling manifold that cannot be used for engine feed shall be scavenged or drained.

* 3.12.1.3 Fuel level control valves. Fuel level control valves shall be designed in accordance with MIL-V-38003. The valves shall be single- or dual-seat valves with dual pilot control devices, except for jettisonable external tanks, where only one control device is required. These pilot control devices shall be level sensing and not operated by sensing tank pressure or by a wetting action, such as thermistors. The valves shall be designed to prevent surge pressure in the refueling manifold from exceeding 120 psig. The valves shall be so located that during all normal ground and aerial refueling attitudes the fuel levels will not exceed the 3 percent normal expansion space allowance of the fuel tanks. When surge relief valves are either in refueling lines or integral with the valve design, the relief setting shall be of a value compatible with the aerial refueling tanker pressure versus flow curves depicted on figure 1. The high level control valve level sensing devices shall be suitably enclosed to prevent foreign objects or fuel tank components from interfering with the level control operation. The valve design shall provide for checking the operation of the primary and secondary pilot control devices in each tank, in the initial portion of the ground refueling operation.

* 3.12.1.4 Float switches. Float switches used in the system shall meet the requirement of MIL-S-25980.

* 3.12.2 Gravity refueling. It shall be possible to gravity refuel the aircraft in 4 minutes or at a minimum rate of 200 gpm. It shall not be necessary to perform any operations other than removing the fuel filler cap and grounding the nozzle prior to refueling. Grounding receptacles shall be provided in accordance with 3.12.1.1.1.

* 3.12.3 Ground defueling. It shall be possible to completely defuel the aircraft through the servicing adapter at a minimum rate of 200 gpm per adapter with a suction pressure at the adapter not to exceed 3 psi. The aircraft fuel boost pumps may be utilized for defueling. In the event of a wheels-up landing, it shall be possible to defuel the aircraft through the normal fuel servicing adapters or by suction through accessible openings in the tank.

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*3.13 Aerial refueling subsystem. The design information contained in the following subparagraphs is applicable for the design of receiver aerial refueling systems. The procuring activity shall be consulted when the design of tanker aerial refueling systems or provisions are being considered. All future aircraft shall have aerial refueling capability installed or the space and structural provisions for aerial refueling. The space and structural provisions for the aerial refueling installation shall be in accordance with the requirements of this specification. Access and appropriate routing holes and support shall be provided for easy installation of all fuel, electrical, and hydraulic lines and control cables. This includes the access holes for the fuel line routing from the fuel tank or single-point refueling line to the point of the aerial refueling installation.

*3.13.1 Receptacle - flying boom. The following requirements apply to production or retrofit installation of boom/receptacle-type aerial refueling systems. The receptacle/slipway shall be located forward of the windshield to provide an adequate view of the refueling operation by the pilot or copilot. The receptacle face shall be a minimum of 2 feet from the windshield. Components such as antennas, lights, etc, protruding above the skin line which would be subject to damage by the boom shall not be located in an area adjacent to the receptacle/slipway. This area is defined as: (1) an 8-foot long by 5-foot wide path forward and centered with the receptacle face, and (2) a 2-foot distance on each side and to the aft of the receptacle face.

3.13.1.1 Angle of receptacle installation. A composite layout shall be prepared to determine the angle at which the receptacle is to be inclined on the receiver aircraft by referencing all normal changes in the receiver's fuselage pitch angle relative to the tanker aircraft's centerline with refueling hardware in the mated configuration. With the smallest receiver fuselage pitch angle at the tanker boom's lower, outer limit (40° elevation, 551 inches extension) as one extreme and the largest receiver pitch angle at the tanker boom's upper, inner limit (20° elevation and 404 inches extension), the angle of installation shall be determined by the following equation:

$$X = 1/2\ (60^\circ - A - B)$$

Where A = minimum fuselage pitch angle during refueling
B = maximum fuselage pitch angle during refueling
X = angle of installation.

The receptacle installation angle as determined by the equation may be slightly adjusted to provide optimum clearance of the tanker boom nozzle with the aircraft structure surrounding the receptacle. The composite layout as required above, the angle calculation, and any other information necessary to determine the receptacle angle of installation shall be approved by the procuring activity.

3.13.1.2 Receiver/tanker performance comparison. A performance comparison of aerial refueling speeds and altitude envelopes of tanker and receiver shall be required. The receiver shall be able to close on the tanker for refueling hookups in all refueling speed ranges without abrupt engine thrust changes such as would be produced by an unmodulated afterburner. Prior to testing, data or calculations illustrating performance compatibility shall be approved by the procuring activity.

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3.13.1.3 Aerial refueling components. All receptacle installations shall incorporate the following components and installation features.

*3.13.1.3.1 Receptacle. The receptacle, including the induction coil and electrical harness and connector, nozzle latches, mounting and trunnion support, toggle contact, toggle latch switch, hydraulic actuator, fuel valve, hydraulic toggle latch actuator, and hydraulic relief valve shall be designed and tested in accordance with MIL-R-27521. The receptacle, mounting provision, and support structure in the aircraft shall be designed to accept the loads as specified in MIL-R-27521. The receptacle shall be located so that the entire boom envelope may be used and so that fuel spray on disconnect will not enter any air intake ducts. The receptacle and slipway shall be located on the centerline of the aircraft. The receptacle/slipway shall be hydraulically actuated with fail-open provisions for emergency refueling. The receptacle/slipway shall move from "closed to open" or "open to closed" within a time period of 5 to 20 seconds under all operating conditions except for the emergency extension mode which shall occur within a maximum of 30 seconds. The receptacle location and installation features shall be approved by the procuring activity before installation of the receptacle.

*3.13.1.3.2 Slipway. A slipway shall be provided to assist the boom operator in inserting the boom. In addition to airloads and other loads normally imposed on the aircraft structure, the slipway and its doors shall be designed for impact loads imposed by contact with the boom nozzle. Ultimate design loads of 2,000 pounds laterally and 5,000 pounds vertically shall be used for the slipway door design. In addition to normal aerodynamic and inertia loads, the aircraft skin backup structure surrounding the receptacle/slipway shall also be designed for impact loads imposed by inadvertent contact with the boom nozzle. Ultimate design loads of 750 pounds laterally and 1,800 pounds vertically shall be used for the skin area 12 inches out from the perimeter of the receptacle face slipway installation. The slipway must be a minimum of 18 inches long, and the sides of the slipway shall be smooth and high enough to guide the nozzle into the receptacle. The nozzle radius of 2-1/2 inches shall be used as a minimum depth for guiding the nozzle into the receptacle. The slipway shall mate with the receptacle and provide smooth transition surfaces. The slipway area shall clear the boom with the maximum receiver pitch angle by a minimum of 5° for any boom angle within the tanker boom automatic disconnect envelope. Slipway design for fighter aircraft may use 2° clearance. The slipway shall have hydraulically operated door(s) which shall be designed to fail in the open position for emergency refueling. Cockpit indication light shall be provided to show when the receptacle/slipway is opened and locked and closed and locked. The design of the slipway and adjacent skin shall be approved by the procuring activity prior to installation.

*3.13.1.3.3 Night refueling lights. The receptacle/slipway installation shall incorporate two flush lights, one on each side of the slipway, to illuminate the slipway and receptacle face for night refueling. The lights shall be directed toward the receptacle bore, and large light angles shall be avoided to prevent boom operator or receiver pilot blinding. The prism lens feature may be utilized. The lights shall meet the requirements of MIL-L-6503, and it shall be possible to dim the lights from the cockpit from full bright to off.

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*3.13.1.3.4 Indicator lights. Indicator lights shall be provided to inform the receiver aircraft pilot of the refueling condition. The lights shall read from top to bottom as follows:

Blue	"Ready" for contact	- This light may also be used to indicate that the receptacle/slipway is locked in the open position. See 3.13.1.3.2.
Green	"Latched"	
Amber	"Disconnect"	

The lights shall incorporate sun shields, transparent lenses with translucent centers, press-to-test switches, and a rheostat. The lights shall be located where the receiver aircraft pilot and copilot may easily observe without changing their visual tanker aircraft reference. The lights shall be so wired that illumination occurs only upon termination of the function being indicated. The indicator lights shall meet the requirements of MIL-L-6503.

*3.13.1.3.5 Amplifier, reset switch, and manual override switch. The signal amplifier used in the aerial refueling system shall be in accordance with MIL-S-38449. A reset switch shall be provided for returning the amplifier to precontact position (ready for contact). The reset switch shall be located near the indicator lights or at a position easily accessible to the receiver aircraft pilot without changing his visual tanker reference. A manual override switch shall be furnished for the signal amplifier to permit operation of the latches for connect and disconnect in the event of an amplifier failure. The override switch shall be located for easy operation by the receiver aircraft pilot without changing his visual tanker reference.

3.13.1.3.6 Disconnect switch. A disconnect switch shall be provided in the aircraft control stick grip for the pilot and copilot. If the copilot position is such that the refueling cannot be conducted from this position, the copilot disconnect switch shall not be required.

*3.13.1.3.7 Fuel pressure disconnect switch. A switch shall be provided for initiating disconnects when manifold pressures indicate a full tank condition or that high level fuel valves have all closed. The switch shall be installed in the fuel line near the receptacle and the settings shall be compatible with high level control valve relief setting and the tanker pump static pressure; that is, a value which will not create inadvertent disconnects.

3.13.1.3.8 Slipway and receptacle drains. Slipway and receptacle drains shall be provided to prevent accumulation of fuel spillage or water in the receptacle/slipway area. Overboard drains shall be located so that drained fuel and water will not enter any air intake ducts.

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3.13.1.4 Retrofit installations. For retrofit installations of aerial refueling provisions on aircraft which do not have fuel level control valves in accordance with 3.12.1.3, the installation of the proper level control valves shall be included in the retrofit.

3.13.1.5 Reverse refueling. For aircraft with a reverse refueling mission requirement, the procuring activity shall be consulted for the specific design requirements for the aircraft.

*3.13.2 Probe drogue. The probe mast and actuating mechanism shall be designed and tested in accordance with MIL-F-8615. The probe mast shall be free of protruding or sharp objects which could catch or damage the drogue. A composite layout shall be prepared to determine the angle at which the probe is inclined on the receiver aircraft by referencing all normal changes in the receiver fuselage pitch angle relative to the tanker aircraft centerline with refueling hardware in the mated configuration. The composite layout, the angle calculations, and any other information necessary to determine the probe angle of installation must be approved by the procuring activity.

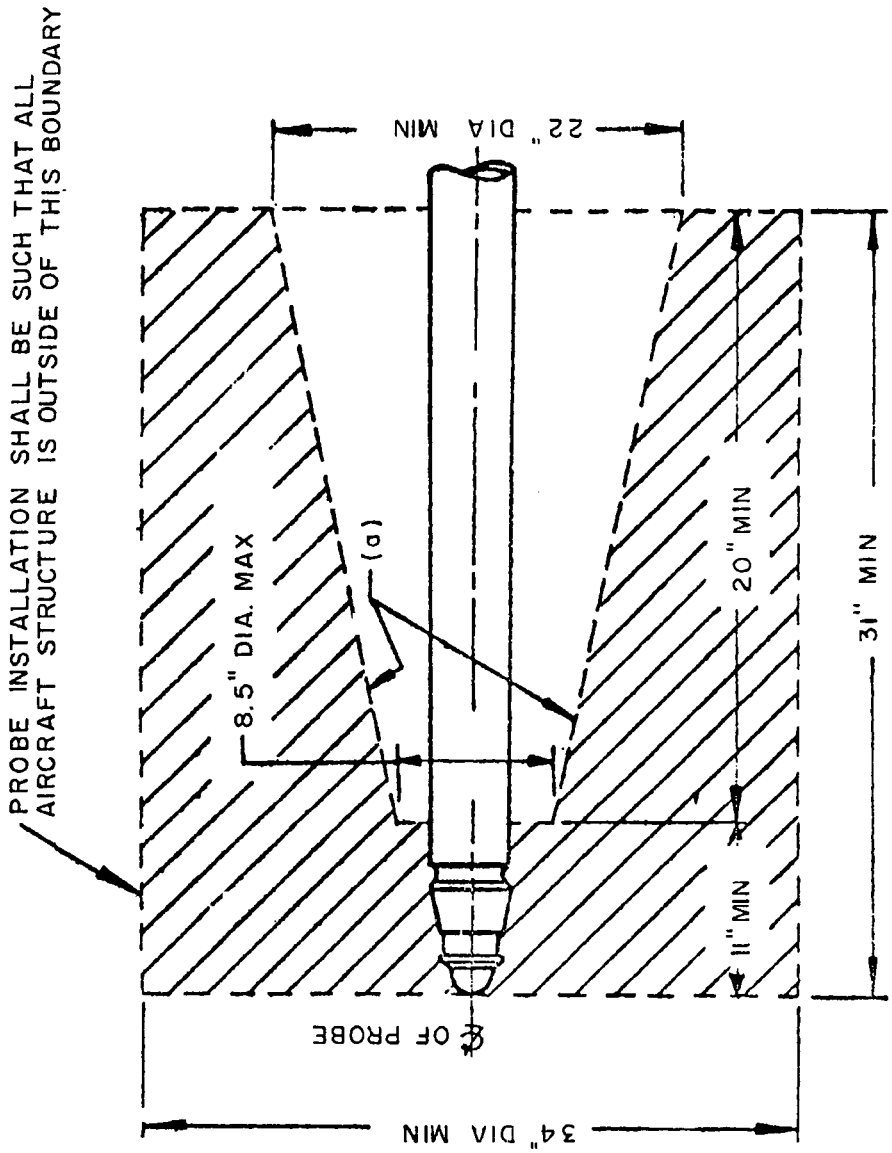
*3.13.2.1 Probe loads for the boom-to-drogue adapter kit refueling. The probe mast and backup aircraft structure shall be designed for ultimate load conditions of: (1) 1,000-pound tensile load in combination with a 3,000-pound radial load, (2) 1,000-pound compression load with a 3,000-pound radial load, (3) a 2,000-pound tensile load, and (4) a 2,000-pound compression load. The tensile load shall be applied at a point equivalent to the latching shoulder parallel to the axis of the Type MA-2 nozzle. The radial load shall be applied to a point equivalent to the nozzle sleeve 3.5 inches from the gage point in the toggle latching groove. The compression load shall be applied at a point equivalent to the lip of the nozzle sleeve and parallel to the longitudinal axis of the nozzle. Limit loads shall be 67 percent of the above ultimate loads. No malfunction or deformation to the probe mast or backup structure shall occur upon completion of each ultimate load test.

*3.13.2.2 Probe loads for reel system refueling. The probe mast and attaching structure shall be designed for ultimate loads of 1,500-pound tension force in combination with a 1,500-pound radial force and 2,000-pound compression force acting separately. Limit loads shall be 67 percent of the above ultimate loads. No malfunction or deformation to the probe mast or backup structure shall occur upon completion of each ultimate load test.

*3.13.2.3 Probe standard parts. Standard parts in accordance with MS24354 shall be used on the probe.

*3.13.2.4 Probe location, envelope, and alignment. The probe shall be located so that the receiver aircraft pilot can easily observe the refueling operation without changing his visual tanker aircraft reference. The pilot shall be able to see a minimum of 12 inches of the probe mast measured from the tip of the nozzle. The probe shall have a minimum clearance with respect to the fuselage in accordance with figure 2. The vertical axis plane of the nozzle shall be parallel to the vertical

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(a) When no other probe mast or aircraft location is available for mounting probe doors, lights, etc , the envelope boundary specified shall be maintained.

(b) With the aerial refueling drogue seated on the probe nozzle there shall be a minimum of 1 foot clearance between the drogue and the windshield of the aircraft.

*FIGURE 2. Probe Clearance Envelope

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axis plane of the aircraft. The longitudinal axis of the nozzle shall be parallel to the line of average fuselage pitch angle. The line of average fuselage pitch angle is a line bisecting the angle between the minimum fuselage pitch angle and maximum fuselage pitch angle during the refueling operation.

3.13.2.5 Externally mounted probe. Any probe mast which cannot be retracted into the fuselage for a flush installation shall be easily removable on the ground. The probe mast shall have the capability of being removed within 15 minutes by a maximum of two men. A drain plug shall be provided for draining the probe mast on removal. Lockwiring of the drain plug shall be provided on installation. The aircraft fuel tube inlet and probe mast outlet shall be sealed on removal of the probe mast.

*3.13.2.6 Retractable probe mast. Normal extension and retraction shall be accomplished by a hydraulic actuator. Emergency backup shall be provided for extending the probe at least one time per flight. The probe must extend or retract within a time period of 5 to 20 seconds under all operating conditions except the emergency extension mode which shall occur within a maximum of 30 seconds. The probe stowage compartment shall be sealed to prevent fuel leakage into other aircraft compartments. The stowage compartment shall be provided with drains in accordance with 3.6.2.8.5 to prevent accumulations of fuel and water.

*3.13.2.7 Telescoping probe mast. The telescoping probe shall have positive mechanical locking and pilot indication of the full extension mode and the fully retracted mode. The probe must telescope or retract within a time period of 5 to 20 seconds under all operating conditions except the emergency extension mode which shall occur within a maximum of 30 seconds.

*3.13.2.8 Lighting. All probe installations shall be provided with a light(s) in accordance with MIL-L-6503 for night refueling. It shall be possible to dim the light(s) from the cockpit, from full bright to off. Prior to contact, the light(s) shall illuminate a minimum of 12 inches of the probe measured from the nozzle tip and the drogue at a distance of 50 yards. After a hookup is established with the drogue of the boom-to-drogue adapter kit, a minimum of 9 feet of hose and 20 feet of the tanker boom shall be illuminated, sufficient for pilot monitoring. The boom operator's visibility shall not be impaired by the receiver's probe light or any other receiver lights during refueling. After a hookup is established with the drogue of a tanker employing an aerial refueling pod, a minimum of 12 feet of hose and the hose exit from the aerial refueling pod shall be illuminated.

*3.13.3 Aerial refueling hardware isolation. A check valve or motor-operated shutoff valve shall be provided to prevent aircraft system pressure on either the boom receptacle or the probe nozzle. When motor-operated valves are employed, they shall incorporate a manual override feature and be accessible during flight. When aircraft employ both the boom receptacle and the probe mast/nozzle, check valves shall be used to isolate each system from the other to prevent recirculation and external fuel leakage during a single failure condition of either the nozzle poppet or the receptacle sliding valve assembly.

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3.14 Fuel jettison subsystem. A fuel jettison subsystem shall be provided for inflight reduction of gross weight.

3.14.1 Fuel jettison rate. The fuel jettison system shall provide a minimum jettison rate per minute equivalent to 1 percent of maximum gross weight of the aircraft, and the aircraft shall be able to jettison fuel to a safe landing weight in a maximum of 10 minutes.

3.14.2 Aircraft controllability. It shall be insured that the fuel jettison operation does not adversely affect the controllability of the aircraft.

3.14.3 Jettison controls. Means shall be provided for closing the jettison valves during flight at any stage of the jettison operation. The fuel jettison control shall be located at the systems engineer's station. In the absence of a systems engineer's station, the control shall be made accessible to the pilot or copilot. The control shall be designed to minimize the possibility of the jettison system being inadvertently or accidentally actuated. The design of the jettison system shall be such that the entire fuel load cannot be dumped unless a second or additional initiating action is taken by the operator.

3.14.4 Fire hazard. The fuel jettison system and its operation shall be free from fire hazard. The fuel shall discharge clear of all parts of the aircraft. It shall be insured that no fuel fumes enter any portion of the aircraft.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. The aircraft fuel system shall be inspected at the mock-up inspection and the inspection of the first complete experimental aircraft by representatives of the procuring activity. The purpose of these inspections shall be to assist in determining compliance with the following requirements of this specification:

a. Design

b. Configuration

c. Installation

d. Fire hazard

e. Any other requirements for which visual inspection will be beneficial in supplementing the contractor's engineering, logistic, or test data.

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* 4.3 Phases of testing. The testing of the fuel system shall be conducted in the following phases:

a. Component testing

b. Fuel system rig testing

c. Ground and flight testing.

4.4 Component testing. Components shall be tested in accordance with the applicable military or contractor specification. Contractor specifications must be approved by the procuring activity prior to testing.

4.5 Fuel system rig testing. The fuel system rig testing shall be conducted in accordance with all the tests specified in 4.8, Rig Test Methods. With approval of the procuring activity, tests may be interchanged between 4.8 and 4.9.

4.6 Ground and flight testing. The ground and flight testing shall be conducted in accordance with 4.9.

4.7 Test conditions

4.7.1 Test fluid. Unless otherwise specified, and approved by the procuring activity, one of the fluids specified in 3.6.2.1 shall be used.

4.7.2 Test rig. The test rig shall be capable of demonstrating fuel system performance under all operational altitudes and attitude of the aircraft. The full scale aircraft fuel system shall be simulated by the test rig. Where practicable, the test rig shall incorporate the actual fuel system components (pumps, valves, piping, connections, etc) that are intended to be utilized in the aircraft fuel system installation. Prior to start of testing, the complete test outline and description of the test rig shall be submitted to the procuring activity for approval.

* 4.7.3 Tolerances. For tests in which a test parameter includes a tolerance, the test shall be conducted at a value of the test parameter as near as possible to the most critical value.

4.8 Rig test methods

4.8.1 Engine feed and transfer subsystem rig tests

* 4.8.1.1 Fuel availability. Rig tests shall be conducted to determine the quantity of fuel available to the engine at sea level engine military rated power conditions utilizing both internal and external tanks by normal feed and alternate methods at the following aircraft attitudes. Fuel remaining in the system subsequent to initial flow fluctuations shall be considered unavailable at normal rated power. The rig shall be refueled at both the ground and aerial refueling attitudes.

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a. Normal ground attitude

b. Takeoff attitude

c. Level flight attitude (landing pattern conditions, low gross weight, low airspeed, low altitude)

d. Landing attitude (touchdown)

e. 10° greater angle than landing attitude

The location of the unavailable fuel shall be specified.

4.8.1.1.1 Internal tank switch-over and fuel transfer rates. Internal fuel tank manual switch-over provisions shall be demonstrated, as applicable. Internal fuel tank transfer rate capabilities shall be demonstrated under conditions considered most critical for the aircraft. Surge pressure resulting from feed and transfer flow fluctuation shall be measured and recorded.

4.8.1.2 Boost and transfer pump pressurization failure tests. Tests shall be conducted to determine the unavailable fuel quantities as a result of various fuel boost pump and transfer pump failures. If internal cell pressurization is used to supplement the boost pump supply system or if cell pressurization constitutes the primary supply system, testing to explore the effect of pressurization failure shall be conducted by disabling the pressurization system in conjunction with appropriate combinations of inoperative boost and transfer pumps, as applicable. The simulated engine(s) shall be operated at military power at normal aircraft flight and landing attitude to:

a. Determine the quantity of unavailable fuel with inoperative boost pumps which normally supply the engine

b. Determine the amount of fuel made unavailable with each fuel transfer pump made inoperative

c. Determine the amount of fuel made unavailable with significant combinations of fuel boost or transfer pumps made inoperative.

4.8.1.2.1 Vapor liquid ratio tests. With a fuel temperature of 110°F and boost pumps inoperative, the vapor liquid ratio forming characteristics of the fuel system shall be determined for military and maximum engine power fuel flow. The vapor liquid ratio shall be measured at 2,000-foot increments from sea level to an altitude which causes the vapor liquid ratio to be in excess of the actual capability of the engine fuel pump. The effects of airframe fuel filter bypassing or impending bypass on system vapor/liquid characteristics shall also be determined. Reid vapor pressure shall be determined for the fuel used during each simulated flight.

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* 4.8.2 Aircraft fuel tank subsystem tests. Two or more internal fuel cells which have a common fuel level through the use of interconnectors may be considered as one tank for the purpose of the tests herein.

4.8.2.1 Fuel tanks

4.8.2.1.1 Tank capacities. Capacity of each fuel tank shall be determined and compared with the design capacity approved by the procuring activity. The tanks shall be filled at the normal ground attitude.

4.8.2.1.2 Sump capacity. Sump capacity of each internal cell at normal ground attitude shall be obtained by measuring the fuel drained from the sump drain after all fuel possible has been pumped from the fuel system by the aircraft boost pumps. The following shall be determined:

a. That the capacity meets applicable design requirements

b. That all cells may be satisfactorily drained on the ground.

Residual fuel, if any, remaining after cells and sumps are drained shall be determined.

* 4.8.2.1.3 Expansion space. The aircraft shall be fueled to normal capacity by pressure and gravity refueling; then the measured fuel shall be added until the expansion space is filled. (Special provisions may be required.) This test shall be conducted to demonstrate that the internal fuel system is provided with the required expansion space at the normal ground attitude without fuel spillage.

4.8.3 Fuel vent subsystem tests

4.8.3.1 Siphoning tests. Tests shall be conducted to determine the siphoning characteristics of the fuel vent system. After normal fueling, the aircraft shall be placed at normal flight attitude and the quantity of fuel, if any, discharged from the fuel vent opening shall be measured. After normal fueling at ground attitude, fuel shall be added until a steady stream of fuel is discharged from the fuel vent opening. The fuel being pumped into the system shall be shut off and the amount discharged from the vent opening, after shut off, shall be recorded.

* 4.8.3.2 Overpressurization. Tests shall be conducted to demonstrate that the vent system is adequate to prevent tank overpressurization as a result of a level control valve failure during simulated ground and aerial refueling conditions. The tests shall include the appropriate range of inlet fuel pressures and flow rates to the receiver's fuel inlet refueling connection as presented by the curves on figure 1. A pressure pickup shall be provided at each vent outlet, in each tank, and between each bladder cell and cell cavity. Values shall be recorded.

4.8.3.3 Fuel tank purging. Purging tests shall be conducted in accordance with MIL-P-5902.

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4.8.4 Fuel quantity gaging subsystem tests

4.8.4.1 Components. The components shall be tested in accordance with sampling plans A and B of MIL-G-26988.

4.8.4.2 System calibration tests. The system calibration tests shall be conducted in accordance with the requirements of MIL-G-7940. These tests shall not be conducted until approval of the engineering report required in MIL-G-7940 has been granted by the procuring activity.

* 4.8.5 Refueling and defueling subsystem tests. The following tests shall conducted on the fuel system test rig to establish conformance to the design requirements.

* 4.8.5.1 Refueling. The time to fill the system shall be recorded using a starting condition as described in 3.12.1.2.b. The flow rates to the system shall be determined for various tank open conditions and for each tank open singly. Tests shall be made for nozzle inlet pressures of 25, 35, 45, and 55 psig for ground refueling and in accordance with figure 1 for aerial refueling.

* 4.8.5.2 Surge pressure test. The refueling manifold shall be instrumented to determine surge pressure encountered during refueling. There shall be a minimum of one pressure pickup at each fuel level control valve and at each ground and aerial refueling connection. The surge pressure shall be measured under the following conditions while refueling through each servicing refueling connection with a pressure of 55 psig for ground refueling and a pressure in accordance with figure 1 for aerial refueling:

a. Maximum flow - All fuel control valves closed simultaneously

b. Maximum flow - Each fuel level control valve closed individually with all remaining fuel level control valves in the closed position

c. Maximum flow - All fuel level control valves closed as a result of filling to capacity.

Surge tests shall also be conducted over the inlet pressure and flow range to the receiver's refueling connection as depicted by curves on figure 1.

* 4.8.5.3 Defueling test. Ground defueling shall be demonstrated. Aircraft internal pumps may be utilized to defuel the system. The time required to defuel the internal cells shall be determined for the maximum discharge flow rate. The maximum flow and fuel remaining in the system after defueling shall be recorded. The pressure in each tank shall be recorded during this test.

4.8.5.4 Fuel servicing nozzle clearance. If a switch or valve actuator has been installed on the adapter, a dummy fuel servicing nozzle incorporating the entire usable envelope allowed by MS29520 shall be fitted to the test rig to determine compliance with 3.12.1.1.3.

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4.8.6 Fuel jettison system tests. Fuel jettisoning tests shall be performed to determine the average jettisoning rate and the quantity of fuel which cannot be jettisoned.

* 4.9 Ground and flight tests. Components shall be qualified or have passed safety of flight tests approved by the procuring activity prior to conducting the ground and flight tests. Each fuel system shall be proof pressure tested prior to flight.

4.9.1 Fuel system icing. The airframe manufacturer shall demonstrate compliance with the requirements of 3.6.2.2.2. The icing test plan shall be approved by the procuring activity prior to testing.

4.9.2 Engine feed and transfer subsystem tests

4.9.2.1 Center of gravity travel. Fuel system center of gravity travel tests shall be conducted for the aircraft normal flight attitude with normal operation of engine fuel feed. The data shall illustrate the effects of all external fuel and auxiliary tank fuel as well as total internal fuel. The aircraft shall be positioned at the normal ground attitude and pressure refueled to normal fuel capacity. The aircraft shall then be positioned to the normal flight attitude and the reactions at scales under the main and nose gears recorded. Fuel shall be removed from the aircraft through all engine feed lines at the normal maximum fuel flow rate down to unavailable fuel. During the fuel pump-out, the reactions at the main and nose gears shall be recorded at 2-minute intervals. In analyzing the data the effects of landing gear position shall be accounted for.

4.9.2.2 Maximum altitude. The maximum altitude at which military power and maximum power operation can be maintained on suction feed shall be determined. The temperature of the fuel and a fuel sample for Reid vapor determinations shall be taken before and after the flights. The temperature of the fuel shall be as close as possible to 110°F immediately before takeoff. The following tests shall be conducted:

a. A normal takeoff and climb to minimum safe altitude shall be performed. All booster and transfer pumps shall be turned off. Multi-engine aircraft may test one engine at a time. The aircraft climb shall be made at military power at best climb speed to service ceiling or until 10 percent power loss occurs or until an objectionable engine surge occurs.

b. The test in item a above shall be repeated using maximum power.

4.9.3 Tank system tests

4.9.3.1 Cell capacities, sump characteristics, and expansion space. The cell capacities, sump characteristics, and expansion space shall be determined on actual aircraft tanks in the same manner as determined in the fuel system rig tests.

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* 4.9.3.2 Tank overpressure test. Each fuel tank on the aircraft shall be tested at pressure values of 1.33 times the maximum pressures as determined in accordance with the requirements of 3.8.1.2. Structural failure, permanent set in excess of that specified on approved drawings, distortion, or fuel leakage shall be cause for rejection.

* 4.9.4 Pressurization and fuel vent system tests. The vent system shall contain a pressure pickup at each vent outlet, in each tank, and between each bladder cell and cell cavity. The pressure data and other data such as rate of climb, rate of dive, altitude, and quantity of fuel shall be recorded in sufficient quantity to demonstrate satisfactory operation of the vent system during all flight maneuvers and during ground and aerial refueling.

4.9.4.1 Fuel and vent system operational tests. Dyed fluid shall be used during the test conditions given below to mark any fuel impingement on the aircraft. The vent system shall be evaluated under the following conditions and maneuvers:

a. The aircraft shall be flown at maximum rate of climb to service ceiling followed by a maximum practical rate of descent dive to minimum safe altitude. The dive shall be accomplished with the minimum safe quantity of fuel aboard.

b. The aircraft shall be flown at maximum engine power at service ceiling and at minimum safe altitude (approximately 5,000 feet) for 10 minutes duration each.

c. A maximum speed run of maximum duration that is most conducive to aerodynamic heating of the fuel, fuel cell, and components shall be conducted.

d. With full fuel tanks it shall be demonstrated that fuel spillage does not occur as a result of takeoff and landing accelerations, afterburner and speedbrake application, and uncoordinated turns.

e. Aircraft in which design performance includes inverted flight or negative g shall be flown in this manner for a maximum period as determined by the model specification at the worst altitude condition.

4.9.4.2 Vent outlet icing tests. The vent outlets shall be observed following the aircraft icing tests to determine the extent of ice accumulation or blockage around the vent outlet. Any ice accumulation shall be photographed for submittal to the procuring activity.

4.9.4.3 Vent system failure tests. Failure of components of the fuel system shall be simulated to cause fuel to vent from the vent outlets. Any impingement of the fuel on the aircraft shall be recorded.

4.9.5 Inerting system tests. Tests shall be conducted to investigate the most critical condition for which the inerting system was designed. Sufficient data shall be recorded to demonstrate satisfactory operation of the system.

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4.9.6 Aerial refueling. The aerial refueling test program shall be conducted in the order listed.

4.9.6.1 Retrofit installations

* 4.9.6.1.1 Proximity flight test. A proximity flight test shall be conducted prior to hardware design and fabrication to determine feasibility, receptacle location and installation angle, or probe location and installation angle. The proximity test shall consist of the following and shall require color motion picture coverage:

a. Evaluate effects of the tanker aerodynamic conditions and down wash on the receiver aircraft during simulated hookups.

b. Determine aerial refueling speed and altitude envelope and if afterburner and speed brakes, trailing-edge flaps, and leading-edge flaps are required for altitude and speed envelope.

c. Instrument the receiver aircraft to measure the fuselage pitch angle while in position behind the tanker during combinations of the following conditions:

(1) Heavy gross weight

(2) Light gross weight

(3) Aerial refueling speed and altitude envelope as determined in item b above

(4) Employ modulated afterburner range as established in item b above, if applicable.

d. Determine suitability of receptacle or probe location by making simulated hookups.

* 4.9.6.1.2 Retrofit installation ground test. The retrofit installation shall be tested in accordance with 4.8.1.1, 4.8.3.2, 4.8.5.1, 4.8.5.2, 4.9.2.1, 4.9.3.2, and 4.9.4.

4.9.6.1.3 Retrofit installation final testing. Retrofit installation final testing shall be the same as the production installation testing.

4.9.6.2 Production installations

* 4.9.6.2.1 Aerial refueling ground tests. The following ground tests shall be conducted prior to the first aerial refueling flight test:

a. Conduct a structural test to limit and then to ultimate load for design conditions specified in 3.13.1.3.1, 3.13.1.3.2, 3.13.2.1, or 3.13.2.2 for receptacle/support structure and fuselage skin or probe/support structure as installed on the aircraft.

b. Perform a complete functional test of all installed aerial refueling components.

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c. Conduct a test to demonstrate that the slipway door/receptacle or probe and probe doors have the ability to extend and retract during hydraulic actuation at -65°F temperatures and icing conditions, if not previously demonstrated during the development phase. Fail-open provisions shall also be demonstrated during above temperature and icing conditions. Time of opening and hydraulic pressures shall be recorded. The actuation shall be accomplished under simulated flight airloads for the most adverse conditions.

d. Conduct a test to demonstrate that the drains have the ability to remove rain and fuel leakage and spillage during all appropriate ground and flight attitudes.

e. (Receptacle installations only.) Evaluate the slipway and transition surfaces from the slipway to the receptacle for possible boom nozzle hang-up areas. Conduct a simulated night lighting test for evaluating the night refueling lights.

f. (Probe installation only.) Evaluate the probe for possible drogue hang-up points and for proper location. Conduct a simulated night lighting test for evaluating the night refueling lights.

g. Conduct a tanker compatibility demonstration using actual tanker aircraft to demonstrate compatibility between tanker and receiver aircraft in respect to flow rates, pressures, and surge pressures.

4.9.6.2.2 Flight test. The following tests shall be conducted as a part of Category I testing. Color motion picture coverage is required for all test flights. Items a and b shall be evaluated for both light and heavy gross weight configurations throughout the speed envelope. The fuselage pitch angle for the receiver aircraft during refueling shall be recorded for both light and heavy weight conditions.

a. (Receptacle installations only.) Explore the entire tanker boom envelope including elevation extension and azimuth in 5° and 6-foot increments for disconnect ability, slipway interferences, and the ability of the tanker boom to tow the receiver aircraft. Hookup envelope shall be established. The test shall be conducted with a boom which has been instrumented to measure loads.

b. (Probe installation only.) Explore the entire tanker hose envelope. Evaluate for possible hose interference, for hookup ability, and disconnect ability. The probe shall be instrumented so that loads resulting from hose flops, off-center disconnects, dead hose of a reel system, and overrun on hookup can be recorded.

c. Determine the optimum receiver aircraft refueling envelope in relation to the following items:

(1) Altitude and speed

(2) Boom extension, azimuth, and elevation angles (for hookup and disconnect)

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(3) Wing flap configuration

(4) Use of modulated afterburner and speed brakes, if required for aerial refueling

d. Determine effect of light-to-moderate air turbulence and moderate-to-heavy air turbulence on the time required to accomplish hookup.

e. Determine adequacy of indicator lights, tanker's receiver or director lights, cockpit switch, and control locations.

f. Evaluate the night lighting adequacy during a transition from twilight to dusk to full dark conditions and also from light to heavy gross weight conditions. The use of the rheostat and any resulting blinding or objectional reflections shall be recorded.

g. Aerial refuel the receiver aircraft to the maximum aircraft gross weight required by mission definition. A minimum of five hookups shall be made with the receiver having a weight of 90 percent of maximum gross weight. The tanker shall record the number of refueling pumps and corresponding flow rates during the refueling. With appropriate instrumentation installed, the receiver shall attempt different valve shutoff combinations specified in 4.8.5.2. Surge, steady-state static pressures, and fill times shall be recorded for each set of test conditions.

h. (Receptacle installations only.) Accomplish the following types of disconnects:

(1) Manifold pressure switch

(2) Normal amplifier system and amplifier override initiated by receiver pilot

(3) Normal and amplifier override initiated by tanker boom operator

(4) All tanker boom automatic envelope disconnects including combinations of left and right azimuth, elevation extremes, and minimum and maximum boom extensions.

i. (Receptacle installations only.) Evaluate emergency refueling such as override features, stiff boom, and fail-safe features. Two brute-force disconnects shall be demonstrated, each performed after a 1-hour flight at 40,000 feet altitude. A minimum of 20 forced disconnects shall be accomplished at all combinations of the already established boom disconnect envelope. This shall be accomplished with an instrumented tanker boom. All tension and compression load data of this paragraph and item h above shall be recorded and shall not exceed the "Tension and Compression Working Loads" listed in the design load requirements under the paragraph entitled "Structural" of MIL-F-27521.

j. (Receptacle installations only.) When reverse aerial refueling is required, conduct reverse flow tests. The tests shall consist of the receiver aircraft pumping fuel to the tanker at predetermined flow rates and pump combinations. A disconnect shall be accomplished during maximum fuel flow conditions.

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4.9.7 Fuel jettison system

4.9.7.1 Fuel impingement. It shall be demonstrated that no fuel or fumes impinge or enter any portion of the aircraft or cause any other unsafe conditions under the following conditions (flaps and gear up) at design gross weight. Dumping rate shall be recorded during the following operation conditions:

a. Climb at the one-engine-inoperative, best rate of climb, with the critical engine inoperative and the remaining engines at maximum continuous power. The critical engine is defined as that engine which, when inoperative, will cause flight conditions such as yaw to be most unfavorable for safe fuel jettisoning.

b. Level flight at 1.4 times the indicated stalling speed, if the results of the previous tests show that this condition could be critical.

4.9.7.2 Controllability. A maximum quantity of fuel shall be jettisoned to determine any adverse effects on the controllability of the aircraft.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable.

6. NOTES

6.1 Intended use. This specification is intended for use by aircraft manufacturers in designing and testing fuel systems for aircraft powered by air-breathing engines.

6.2 Definitions

6.2.1 Available fuel. Available fuel is fuel which can be pumped from the aircraft when the tanks have been filled to automatic shutoff at the normal ground refueling attitude subsequent to initial flow fluctuations.

6.2.2 Unavailable fuel. Unavailable fuel is fuel remaining in the aircraft after the available fuel has been removed.

6.2.3 Sump fuel. Sump fuel is fuel which can be drained from the aircraft sumps after the available fuel has been removed from the aircraft.

6.2.4 Trapped fuel. Trapped fuel is fuel remaining in the aircraft after the sump fuel has been drained.

6.2.5 Slipway. The slipway is the specially designed "beefed-up" fuselage skin area forward of the receptacle which serves as a guide to assist the boom operator in hookup operations. The use of this area is required to stabilize the boom nozzle in the vertical and horizontal planes due to the relative motions which exist prior to contact and hookup. Relative motions which exist between the receiver and boom are

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normally produced; when the aircraft are subjected to weather turbulence, boom hookups attempted at lower boom elevations (35° to 40°), out-of-rig booms, and inherent boom instability.

6.2.6 Brute-force disconnect. Brute-force disconnect is the process of disconnecting against the force of locked receptacle toggle latches.

6.2.7 Receiver director lights. Two rows of lights are contained on the bottom fuselage forward of the wheel well of the KC-135 tanker. These lights give location information to direct the receiver aircraft pilot in the forward-aft and up-down boom envelope.

6.2.8 Reverse refueling. Reverse refueling is the process of transferring fuel from the receiver aircraft to the tanker aircraft.

* 6.3 International standardization agreement. Certain provisions (identified by paragraphs below) of this specification are the subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Paragraph No.	Standard
3.8.1.10	ABC Air STD 17/11
3.8.1.10	STANAG No. 3212
3.8.1.10	STANAG No. 3294
3.12.1	ASCC Air STD 11/3
3.12.1.1.2	ABC Air STD 11/2
3.12.3	ABC Air STD 11/6
3.13.2.3	ABC Air STD 17/29

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Review activities:
Air Force - 70, 82

User activities:
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Preparing activity:
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APPENDIX

10. SCOPE

10.1 This appendix lists the data necessary for the procuring activity to properly evaluate a developmental fuel system.

*

20. REQUIREMENTS

20.1 Data and reports. The following data and reports are necessary to thoroughly define the fuel system for the procuring activity:

a. Engineering Design Analysis Report. This report shall include the following:

(1) Result of trade studies performed to arrive at the basic fuel system configuration

(2) Temperature calculation showing maximum fuel temperature resulting from aerodynamic/thermodynamic heating of the fuel and maximum ambient air temperature

(3) Calculated system performance

(4) Predicted problem areas with proposed method of solution

(5) Development test program

(6) Special facilities required for testing of fuel system

* (7) Deviations from this specification and reason for deviation.

b. Fuel System Functional Analysis Report. This report shall include the following:

(1) Description of fuel system

(2) Fuel system schematic diagrams identifying all components, lines, sizes, and relative location

(3) List of Qualified Products List (QPL) components giving name and location of component, manufacturer's part number, and vendor's part number

(4) List of off-the-shelf components giving name and location of component, manufacturer's part number, vendor's part number, applicable military specification, approval status, procurement or model specification, and past usage

(5) List of critical components giving name and location of component, manufacturer's part number, vendor's part number, detail drawings, applicable military specification, procurement specification, and test report

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(6) Fuel system wiring schematics from distribution bus to all fuel system components

(7) Perspective or isometric drawing illustrating all major components and fuel lines in relation to the airframe structure

(8) Outline of fuel scheduling and management

(9) Failure analysis describing the effect of each possible single failure on the operation of the system or subsystem

(10) Fuel system passive defense analysis

* (11) Procuring activity approval items required by paragraphs in this specification:

(a) 3.6.2.8.1.4
(b) 3.7.8.1
(c) 3.11.4
(d) 3.13.1.1
(e) 3.13.1.2
(f) 3.13.1.3.1
(g) 3.13.1.3.2
(h) 3.13.2
(i) 4.4
(j) 4.5
(k) 4.7.1
(l) 4.7.2
(m) 4.8.4.2
(n) 4.9
(o) 4.9.1

c. Systems Test Plan. Outline of proposed test program.

* d. Systems Test Report. Results of system tests required in 4.8 and 4.9 of this specification, motion pictures of aerial refueling flight tests, and photographs required in 4.9.4.2.

e. Fuel System Detailed Weight Statement. The following weight statement shall be submitted as complete as possible during the proposal stage. The weight statement should be up-dated for design reviews as data becomes available.

Fuel System Detailed Weight Statement:

ITEM	WEIGHT
(1) Tanks	
(2) Integral Tank Seals and Sealant	

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(3) Backing Board

(4) Tank Supports and Padding

(5) Tank Bay Sealing

(6) Tank Release and Controls

(7) Fuel Quantity Gaging System

(8) Fuel Heating and Controls

(9) Fuel Vapor Recovery

(10) Fuel Vapor Inerting System and Suppression
Cycle and Supports

Generators

Controls, Etc

(11) Pump Installation
Engine Driven

Booster

Hand (Including Controls)

Transfer

Electrical System (Including Controls)

Pump Supports

(12) Ground Fueling System
Plumbing

Controls

(13) Aerial Refueling Hardware
Plumbing

Controls

Additional Structure

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(14) Engine Feed and Transfer System

Plumbing

Controls

(15) Vent System

(16) Pressurization System

(17) Jettison System

(18) Fuel Drain System

TOTAL

f. System Detail Design Data. The applicable portion of the following list shall be submitted as complete as possible during the proposal stage. The list should be updated for design reviews as data becomes available.

(1) Fuel: Primary Alternate

Consumption at takeoff (lb/hr/engine)

Consumption at cruise (lb/hr/engine)

(2) Tanks, Fixed: (Integral, Bladder, Nonintegral, External)

Specification

Type

Number Quantity/Tanks

Location (Station/WL/BL)

Quantity

Design Crash (g/capacity/angle)

Burst (psig)

Negative Pressure Capability (psig)

Overall Dimensions (in.)

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Largest Opening/Location (in./station)

Sump Location/Capacity (station/gal)

Expansion Space (gal)

Drain Location

Baffles Axial/Transverse (location and number)

Supplier

(3) Fuel Capacity, Fixed:

Total Fixed Capacity (gal)

Trapped Main Fuel (gal)

Usable Main Fuel (gal)

Sump Fuel (qt)

Expansion Space (gal)

(4) Tanks, Auxiliary: (Internal, External) Same as (2).

(5) Fuel Capacity, Auxiliary:

Total Auxiliary Capacity (gal)

Trapped Auxiliary Fuel (gal)

Usable Auxiliary Fuel (gal)

Sump Fuel (gal)

Expansion Space (Auxiliary) (gal)

(6) Tank Materials:

Integral Structure

Specification

Heat-Treat/Grade

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Sealants

Specification

Method of Application

Bladder Support Structure

Bladder, Self-Seal

Specification

Description

Bladder, Nonseal

Specification

Description

Nonintegral Internal Tank Structure

Specification

Heat-Treat/Grade

Ultimate Tensile Strength

(7) Refuel:

Gravity Fill

Drain Line Size (in.)

Drain Discharge Point

Filler Locations

Maximum Fill Rate (gpm)

Single Point

Location

Receptacle Type/Standard

Rated Flow/Pressure (gpm/psig)

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Level Control Valve Pressure Check Method

Line Size (in.)

Burst Pressure (psig)

Fill Sequence

Aerial Fill

Method

Location (station/WL/BL)

Rated Flow/Pressure (gpm/psig)

Line Size (in.)

Fill Sequence

(8) Defuel:

Fixed Tanks Defuel Subsystem

Description

Location (station/WL/BL)

Receptacle Type/Standard

Line Size (in.)

Rated Flow/Pressure (gpm/psig)

Auxiliary Tanks Defuel Subsystem

Description

Location (station/WL/BL)

Receptacle Type/Standard

Line Size

Rated Flow/Pressure (gpm/psig)

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(9) Jettison:

Description

Location (station/WL/BL)

Line Size (in.)

Rated Flow/Pressure (gpm/psig)

Pressure Source

Method of Low Quantity Cutoff

(10) Vents:

Antisyphon Design Decription

Outlet Locations (station/WL/BL)

Size of Line (in.)

Lightning Protection

Special Design Feature

(11) Negative G Provisions:

Type Description

Time at Military Power (min-sec)

Time at Cruise Power (min-sec)

(12) Center of Gravity Control:

Tank Usage Sequence

Normal Center of Gravity Travel

Abnormal Center of Gravity Travel Possibilities

Location and Type of Fuel Management Controls

(13) Critical Components:

Number and Description of Surveillance Items

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Number and Description of Single Failure Critical Items

(14) Boost Pumps:

Location (station/WL/BL)

Power

Circuit Protection

Flow/Pressure at TO Power

Flow/Pressure at Cruise

Type

Speed

Supplier

Special Features

(15) Other Pumps: (Transfer Jettison, Ejection, etc)

Function

Type of Pump

Location

Power

Circuit Protection

Speed

Flow/Pressure in Operation (gpm/psig)

(16) Strainers:

Type

Overall Dimensions

Location

Inlet Size

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Outlet Size

P at TO Flow/60°F

P at Bypass/60°F

Heater Power

Mesh or Micron Rating

Supplier

(17) Plumbing:

Hose Specification

Fitting Material

Flex Coupling Type

Rigid Line Material

Wall Thickness

Ultimate Tensile Strength

Design Burst Factor

Fitting Type

Line Sizes for:

Vents

Crossfeed

Jettison

Single-Point Refueling

(18) Quantity Instrumentation:

Type

Overall Indicator Size

Switch Positions

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Probe Location

Low Level Locations

Low Level Quantity

Supplier

(19) Flow Instrumentation:

Type

Sensor Location

Indicator Ranges

Accuracy

Readout Data

Supplier

(20) Pressure Instrumentation:

Type

Function

Sensor Location

Indicator Ranges

Actuation Pressure

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